HW_3

Quinn Suligoy

1

a

Let x be a student
Let $P(x)$ be the statement "x has taken a political science class"
Let $Q(x)$ be the statement "x has read *The Clash of Civilizations and the Remaking of World Order*"
$\forall x(P(x) \rightarrow Q(x))$

b

Let x be an integer
Let $P(x)$ be the statement "x is an even number"
Let $Q(x)$ be the statement "x is a prime number"
$\exists x(P(x) \wedge Q(x))$

2

a

Let x be a person
Let $P(x)$ be the statement "x has registered to vote"
Let $Q(x)$ be the statement "x is under 14 years old"
$\exists x(P(x) \wedge Q(x))$

b

Let x be an integer
Let $P(x)$ be the statement "x is greater than 0"
Let $Q(x)$ be the statement "$3x - 5 > 0$"
$\forall x(P(x) \rightarrow Q(x))$

3

a

Let x be a person

Let $P(x)$ be the statement "x needs somebody to love"
$\forall x(P(x))$

b

Let x be a person

Let $P(x)$ be the statement "x lives in Australia"

Let $Q(x)$ be the statement "x has visited some country in Europe"
$\forall x(P(x) \to Q(x))$

4

a

Let x be a person

Let $P(x)$ be the statement "x lives in the United States"

Let $Q(x)$ be the statement "x has a relative in another country"
$\forall x(P(x) \to Q(x))$

b

Let x, y be a person

Let $P(x, y)$ be the statement "x and y have birthdays in January"

Let $Q(x, y)$ be the statement "x and y are younger than 20 years old"

Let $R(x, y)$ be the statement "x is not the same person as y"
$\exists x, y(P(x, y) \land Q(x, y) \land R(x, y))$

5

a

$\exists x \exists y((x > 2) \land (x < 6) \land (y > 5) \land (y < 8) \land (x^2 + y^3 = 280))$
possible x values: 3,4,5

possible y values: 6,7

The range of the function is the values:
$3^2 + 6^3 = 9 + 216 = 225$
$4^2 + 6^3 = 16 + 216 = 232$
$5^2 + 6^3 = 25 + 216 = 241$
$3^2 + 7^3 = 9 + 343 = 352$
$4^2 + 7^3 = 16 + 343 = 359$
$5^2 + 7^3 = 25 + 343 = 368$
Which does not include 280. Therefore the statement is false

b

Take the equation $x + y = 2z$ we can construct a general z value through algebra, $z = \frac{x}{2} + \frac{y}{2}$. Then for every integer where $x > 11$ and $y > 200$ there does exists a z value to satisfy the statement, making it true.

6

a

$(4, 10)$ is a lower bound for the function and $4^2 + 10^2 = 116 > 115$ therefore there cannot be any values in the domain of the function that satisfy it. Making the statement false.

b

$y = 20$ is an upper bound for the function. Now consider the statement $(-0.1 > -1) \rightarrow (-0.1 + y^2 \geq 400)$, plugging in our upper bound we get $(-0.1 > 1) \rightarrow (399.9 \geq 400)$ this statement evaluates to false. Therefore the original statement is false as for $x = -0.1$ there exists no y.

7

4b was the most difficult for me as it involved needing two variables rather than one.